

BSSM4GSQ code package

Michael E Sparks

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1 Getting Started

This software facilitates generation of splice site probabilities for use with **GeneSeqer**, and training data with which to build splice site parameters using the **gthbssmbuild** tool in the **GenomeThreader** package. **BSSM4GSQ** can process either plain text **GeneSeqer** or gthXML v1.0 (or later) output. gthXML output can be produced natively by the **gth** program of **GenomeThreader**, and can be produced from plain text **GeneSeqer** output using the **GSQ2XML.pl** script, available from either <http://www.genomethreader.org> or the gthXML-tools repository at <http://brendelgroup.org/mespar1/gthxml/>. The end user may wish to study the following reports prior to using this software:

1. Brendel V, Xing L, Zhu W. (2004) Gene structure prediction from consensus spliced alignment of multiple ESTs matching the same genomic locus. *Bioinformatics*. **20**:1157-69.
2. Sparks ME and Brendel V. (2005) Incorporation of splice site probability models for non-canonical introns improves gene structure prediction in plants. *Bioinformatics*. **21**:iii20-iii30.

2 Directions

1. Verify that the following executables are present in the bin/ directory.
 - (a) **indexFasSeq**
 - (b) **BSSM_build**
 - (c) **BSSM_print**

If any of these are absent, `cd` to the `src/` directory and issue “`make`”. (This step is, however, optional, as the `Mktraindata.sh` and `Mkbssmparm.sh` scripts will build the files, when necessary.)

2. There are two subdirectories in the input directory, `gsq/` and `fas/`. You absolutely must meet the following requirements to make the code work.
 - (a) There must be a one-to-one correspondence between files in these two directories.
 - (b) Files in the `fas/` directory must have an extension of “`.fas`” and those in the `gsq/` directory must have one of either “`.gsq`” or “`.xml`”, for plain text **GeneSeqer** or **gthXML** formatted data files, respectively. You cannot mix plain text and **gthXML** input files.
 - (c) The basename prefixes of cognate `fas/gsq` file pairs, i.e., the substrings prior to “`.gsq`” or “`.fas`”, must be identical.
 - (d) All references made to a genomic template in the spliced alignment output file must be identical to the file’s “file handle” mentioned above. This essentially mandates that each `fas/gsq` cognate file pair correspond to one genomic sequence and its spliced alignment annotation, respectively.

There are sample training data in the `input/sample/` directory. These data will not generate any meaningful probabilities, and are only intended to demo the system.

3. Edit `Mktraindata.sh` such that the `FORMAT` variable is set correctly for the files placed in the `input/gsq/` directory; this is described explicitly in the header of the script. Run `Mktraindata.sh`. This produces exon and intron data, sorted according to phase and placed in the `output/exons_introns/` directory, and sampled, phase-sorted BSSM training data placed in the `output/training_data/` directory. In each of these directories, data will be written to a subdirectory named according to the donor/acceptor dinucleotide termini trained for. (If this is unclear, inspect the contents of the output directories after unpacking this code, run the script, and look at them again.)

`Mktraindata.sh` processes GT-AG introns by default. For other types, tune the `DON` and `ACC` variables (set these in CAPITAL letters!) and run it again. This will not overwrite any existing output in

the `training_data/` or `exons_introns/` directories so long as a different DON/ACC combination is used. Rerunning the script using a DON/ACC pair whose results were already recorded will cause the original data to be overwritten.

4. Run `Mkbssmparm.sh`. The script will solicit some configuration information:

- (a) Name of output file (“foo.bssm”)
- (b) Root directory of training data. (If you haven’t done anything non-standard up to this point, it should be safe to just say “y” here.)
- (c) Build GT model? For GT-AG parameterizations. (If you trained for these intron types, responding with “y” will put the probabilities in your *.bssm file. Else, say “n”.)
- (d) Build GC model? For GC-AG parameterizations. (If you trained for these intron types, responding with “y” will put the probabilities in your *.bssm file. Else, say “n”.)
- (e) File to write ascii data to (“foo.bssm.ascii”)

(Users of the **GenomeThreader** package (<http://www.genomethreader.org>) should note that these binary *.bssm files are **not compatible** with those used by that system. Code in **BSSM4GSQ** implementing the weight array matrix development routines was adapted to implement the **gthbssmbuild** program of **GenomeThreader**; given input data like that produced in steps 1-3 above, **gthbssmbuild** will generate **GenomeThreader-compatible** binary *.bssm files. Please refer to the **GenomeThreader** manual or contact Gordon Gremme at gordon@gremme.org for details.)

The `BSSM_print` utility allows the user to generate an ascii representation of the trained splice site probability matrices. The *.bssm.ascii file presents the weight array matrices in the following order:

```
for TERMINAL in (0-1):
  for HYPOTHESIS in (0-6):
    print transition probabilities
```

where for TERMINAL, 0 and 1 index donor and acceptor sites, respectively; and for HYPOTHESIS, 0, 1, 2, 3, 4, 5, 6 index the T1, T2, T0, F1, F2, F0 and Fi hypotheses, respectively.

The user will, unfortunately, have to manually splice the new probabilities into the daPbm7.* header files included with the **GeneSeqer** source distribution (adjust the GU_7/GC_7 and AG_7 arrays, the name_model array, and the NMDLS macro accordingly) to incorporate the models into the software. There is a script provided in the src/plscripts/ directory, **punctuation.pl**, that will assist in adding syntactical markup to allow copying/pasting results into the header files, e.g.,

```
$ cat something.bssm.ascii | ./punctuation.pl
```

You can verify that your parameter file contains valid probability mass distributions by using the **verify_pmf.pl** script in the src/plscripts/ directory, e.g.,

```
$ cat something.bssm.ascii | ./verify_pmf.pl
```

Please see the commentary in that file for more details; output of a row of zeros is expected, and does not signal an error.

3 Contact Info

If you have questions or concerns, please email me at mespar1@gmail.com.